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FEET
THE SPAN
OF THE
BROOKLYN
BRIDGE

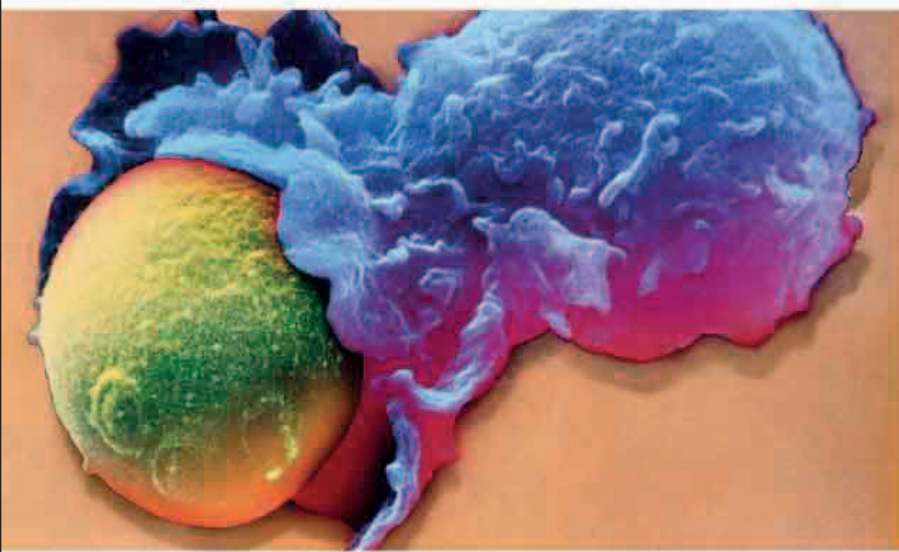
The Brooklyn Bridge spans the East River between Brooklyn and the island of Manhattan in New York City. It was the longest bridge in the world when it was built.

BUILDING ON the work of British surgeon Joseph Lister and French microbiologist Louis Pasteur (see 1870–71), German physician **Robert Koch** (1843–1910) **isolated the organism that causes tuberculosis** (TB) in 1882. He discovered that TB was transmitted in water droplets and could quickly spread in overcrowded slums.

Another medical landmark was achieved when Russian biologist **Élie Metchnikoff** (1845–1916) discovered **phagocytosis**—the process used by the immune system to remove bacterial invaders. In phagocytosis, one cell engulfs another and, in effect, eats it. This discovery led to an **improved understanding of the immune system**.

“ I CONCLUDE THAT THESE **TUBERCLE BACILLI** OCCUR IN ALL TUBERCULOUS DISORDERS, AND THAT THEY ARE **DISTINGUISHABLE** FROM ALL OTHER **MICROORGANISMS**. ”

Robert Koch, German physician, from *The Etiology of Tuberculosis*, 1882

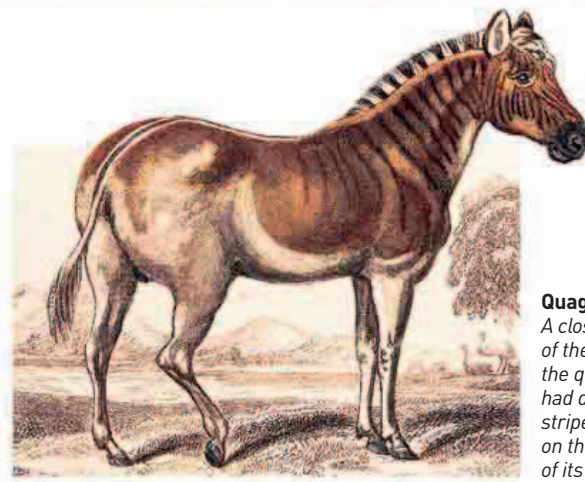


German botanist **Eduard Strasburger** (1844–1912) also contributed to the understanding of how cells work. He coined the terms **cytoplasm** for the jellylike outer region of a cell, and **nucleoplasm** for the compact material of the cell nucleus.

In 1882, Italian volcanologist and meteorologist **Luigi Palmieri** (1807–96) made the **first observation of the element helium** on Earth by conducting spectral analysis (see panel, opposite) of lava during an eruption at Mount Vesuvius, Italy. Previously, the element had only been identified by analysis of light from the Sun.

The following year, English polymath **Francis Galton** came up with the **controversial concept of eugenics**. His theory aimed to improve the human race by selective

Phagocytosis
In this color-enhanced micrograph, a lymphocyte (white blood cell) engulfs a yeast cell in the process known as phagocytosis.



Quagga
A close relative of the zebra, the quagga had distinctive stripes only on the front of its body.

breeding, and was later misused by the Nazis as an excuse for their attempted extermination of the Jews. The scientific basis for eugenics drew in part on the germ line theory developed by **August Weismann** (1834–1914), which stated that characteristics were only passed on by egg and sperm cells, and were not affected by other cells of the body (somatic cells). So, for example, a bodybuilder who develops muscles through exercise will not pass on the muscles to his children. This **marked the end of Lamarckism**, according to which acquired characteristics could be passed on to children (see 1809).

A more practical contribution to human society was made by **Osborne Reynolds** (1842–1912), an Irish-born engineer who

studied fluids. In 1883, he came up with what is now known as the **Reynolds number**, which characterizes the way fluids flow. Today, Reynold's work is important in the designing of pipes to carry different fluids, and in shipbuilding, where the behavior of full-size vessels must be estimated from models tested in water tanks.

On May 24, 1883, the longest bridge of its time, the **Brooklyn Bridge**, was opened in New York City. It was the world's first steel suspension bridge.

A female **quagga**, the last surviving member of its species, died in a zoo in Amsterdam in 1883. This southern African species had been extinct in the wild since the late 1870s.

March 24, 1882
Robert Koch announces his discovery of the tuberculosis bacterium

1882 Élie Metchnikoff discovers phagocytosis

1882 Eduard Strasburger coins the terms cytoplasm and nucleoplasm

1882 Luigi Palmieri detects helium on Earth for the first time

1882 Electric light is introduced in New York City, USA

1883 Francis Galton coins the term eugenics

1883 Osborne Reynolds publishes a paper on the flow of fluids, describing Reynold's number

May 24, 1883
The Brooklyn Bridge opens in New York City, USA

1883 German zoologist Wilhelm Roux suggests that chromosomes carry hereditary factors

August 12, 1883 The last quagga dies in a zoo in Amsterdam